

Signals & Systems

In-Sem Exam • Easy-to-Read Quick Revision Notes

Priority topics: elementary signals, classification, periodicity, even/odd decomposition, system properties, LTI systems, convolution, block diagrams, energy & power.

1. Elementary Signals

Unit Step Signal

$$u(t) = 1 \text{ for } t \geq 0; u(t) = 0 \text{ for } t < 0$$

Sketch: a step that jumps from 0 to 1 at $t = 0$.

Example: $u(t - 2)$ starts at $t = 2$.

Unit Impulse Signal

$$\delta(t) = 0 \text{ for } t \neq 0, \text{ and } \int_{-\infty}^{\infty} \delta(t) dt = 1$$

Important: Sifting property: $x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$.

Sketch: an arrow at $t = 0$ representing unit area.

Ramp Signal

$$r(t) = t u(t)$$

Sketch: zero for $t < 0$; straight line increasing from the origin for $t \geq 0$.

Exponential Signal

$$x(t) = A e^{at}$$

- $a > 0 \rightarrow$ growing exponential
- $a < 0 \rightarrow$ decaying exponential
- Discrete form: $x(n) = a^n$

Sinusoidal Signal

$$x(t) = A \sin(\omega t + \phi)$$

- A = amplitude
- ω = angular frequency
- ϕ = phase
- Period: $T = 2\pi/\omega$

2. Classification of Signals

Type	Meaning / Example
Continuous-time	Defined for every real t ; e.g. $x(t) = \cos t$
Discrete-time	Defined at integer n ; e.g. $x(n) = n^2$
Deterministic	Completely predictable; e.g. $\sin t$
Random	Cannot be predicted exactly; e.g. noise
Even	$x(t) = x(-t)$; symmetric about y-axis
Odd	$x(t) = -x(-t)$; symmetric about origin

3. Periodicity

Continuous-time: A signal is periodic if $x(t) = x(t + T)$, where T is the fundamental period.

For sinusoids: $T = 2\pi/\omega$

Example: $x(t) = \sin(4t) \rightarrow T = 2\pi/4 = \pi/2$.

Discrete-time: $x(n) = x(n + N)$. For $\cos(2\pi n/N)$, the period is N .

Sum of periodic signals: find the LCM of the individual integer periods when applicable.

Example: $\cos(2\pi n/4) + \sin(2\pi n/6) \rightarrow N = \text{LCM}(4,6) = 12$.

4. Even and Odd Decomposition

Every signal can be represented as $x(t) = x_e(t) + x_o(t)$.

Even part: $x_e(t) = [x(t) + x(-t)] / 2$

Odd part: $x_o(t) = [x(t) - x(-t)] / 2$

Example: $x(t) = e^t \rightarrow x(-t) = e^{-t}$

$$x_e(t) = (e^t + e^{-t})/2$$

$$x_o(t) = (e^t - e^{-t})/2$$

5. System Properties

A system transforms an input signal into an output: $y(t) = T\{x(t)\}$.

Property	How to check	Quick clue
Linearity	$T\{ax + bx\} = aT\{x\} + bT\{x\}$	Constant addition often makes it nonlinear
Time invariance	Shift input $\rightarrow$ output must shift by same amount	Compare $T\{x(t-t_0)\}$ with $y(t-t_0)$
Causality	Output uses only present/past input	$x(t+1) \rightarrow$ non-causal
BIBO stability	Bounded input must produce bounded output	Finite gain systems are often stable
Memoryless/static	Output depends only on input at same time	$y(t) = 2x(t)$
Dynamic	Output depends on other time instants	$y(t) = x(t-1)$

Example 1: $y(t) = 2x(t) + 5 \rightarrow$ nonlinear because the constant +5 violates homogeneity.

Example 2: $y(t) = x(t+1) \rightarrow$ non-causal because it uses future input.

Example 3: $y(t) = x(t-3) \rightarrow$ time invariant.

6. LTI Systems and Impulse Response

LTI = Linear + Time Invariant. The impulse response $h(t)$ completely characterizes an LTI system.

For an input $x(t)$, output $y(t)$ is the convolution of $x(t)$ and $h(t)$.

$$y(t) = x(t) * h(t)$$

7. Convolution

Continuous-time convolution:

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

Discrete-time convolution:

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

Graphical steps: 1. Flip one signal $\rightarrow$ 2. Shift $\rightarrow$ 3. Multiply overlapping portions $\rightarrow$ 4. Integrate/sum.

Common result: $u(t) * u(t) = t u(t)$.

8. Block Diagram Representation

Block diagrams represent mathematical systems using **adders, gains, and integrators** (or differentiators where appropriate).

Typical flow: Input → Gain / Integrator blocks → Summing point → Output.

For differential equations, rearrange the equation so the highest derivative is isolated, then implement successive integrations to obtain lower-order states.

Exam tip: clearly label every gain, integrator, summing point, input and output.

9. Energy and Power

Energy of a continuous-time signal:

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Energy signal: $0 < E < \infty$.

Average power:

$$P = \lim_{T \rightarrow \infty} [1/(2T)] \int_{-T}^T |x(t)|^2 dt$$

Power signal: finite, non-zero average power with infinite total energy.

Rule of thumb: finite-duration decaying signals are commonly energy signals; non-decaying periodic signals are commonly power signals.

10. One-Page Formula Sheet

Topic	Formula
Unit step	$u(t)=1$ for $t \geq 0$; 0 for $t < 0$
Impulse	$\int \delta(t) dt = 1$
Ramp	$r(t) = t u(t)$
Sinusoid period	$T = 2\pi/\omega$
Even part	$x_{\text{even}}(t) = [x(t) + x(-t)]/2$
Odd part	$x_{\text{odd}}(t) = [x(t) - x(-t)]/2$
CT convolution	$y(t) = \int x(\tau)h(t-\tau)d\tau$
DT convolution	$y(n) = \sum x(k)h(n-k)$
Energy	$E = \int x(t) ^2 dt$
Power	$P = \lim_{T \rightarrow \infty} (1/2T) \int x(t) ^2 dt$

Last-minute exam priority: 1) Signal sketches → 2) Periodicity → 3) System properties → 4) Even/odd decomposition → 5) Convolution/LTI → 6) Energy & power.

Exam writing tip: Always write the governing formula first, substitute values step-by-step, and draw a small labeled sketch wherever the question involves a signal.